

Fortessa Standard Operating Procedure & User Guide - STARTUP
UC Davis Flow Cytometry Shared Resource

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1. Power on the Fortessa:
 - a. Wait 15seconds then proceed to the next step
 - b. Power on the UV laser – OPSL software: You will need to click the power switch to turn off the laser (it defaults to “ON” and this always causes a fault, this is normal), once the switch is in the “OFF” position type 60 into the power level box and turn the laser on once more. A green power light should blink in the interface. Once this is solid it should energize and works its power level up to 58.XXmW.

Note: The lasers on the Fortessa require 15-20min of warm up time.

2. While the system is warming up check all liquid levels
 - a. Check that the sheath is filled to the weld line on the tank
 - b. Check that the waste is less than half full – you may need to empty and fill with 1L bleach if you have a lot of samples to run.
3. Purge air bubbles from system
 - a. Obtain the empty beaker
 - b. *Purge air from sheath filter*
 - i. Lightly tap the filter on the floor or sheath cube at a 45 degree angle in order to move all air bubbles to the filter evacuation tube/port. Open the relief valve and allow sheath to flow into the beaker for 5seconds or until all air is purged from the filter then close the relief valve.
 - c. *Purge air from sheath tubing*
 - i. Move to right side of instrument and open the sheath tubing relief value and allow sheath to flow into the beaker for 10seconds, close the relief valve.
 - d. *Purge air from inside system*
 - i. Remove the water tube from the instrument SIP (leave the tube holder to the side!)
 - ii. Press the “Prime” button and wait for the light to return to the standby button, this is one prime cycle. Do this a total of 3 times.
 - e. *Purge air from waste tubing*
 - i. Replace the tube of DI water on to the SIP.
 - ii. Confirm the instrument is set to ‘Low” and press the “Run” button.
 - iii. Watch the middle waste line and ensure all air is pushed into the waste tank and the entire waste line is a solid column of liquid. *note: sometimes this will require flicking the waste line as the bubbles can get caught
 - iv. Press the instrument [STBY] button to put the system into StandBy..
4. Launch Diva and Log into the software

5. Diva will now try to connect to the Fortessa, this could take upwards of 5min. Note: -connecting to cytometer- should be visible on the bottom of the “cytometer window” and in the bottom right of the entire Diva workspace window.
 - a. *note: on the rare occasion Diva and the instrument do not connect a complete restart of both the instrument and the computer are required.
6. Once connected a dialog box will often popup stating a mismatch between current settings and CST always select “use CST settings”
 - a. *note: On occasion this box will NOT pop up on first log in, this is not a problem as the message will appear after running CST.
7. QC the instrument
 - a. Select **Cytometer** → **CST** from the top menu
 - b. Allow CST to connect to the instrument. *note: Diva is now locked to user input as the instrument is now connected with CST
 - c. Remove air from droplet containment system
 - i. Ensure a tube of DI water continuing at least 2mL is on the Fortessa SIT
 - ii. Move the SIT arm to the side and listen for the drop containment pump.
 - iii. Allow 1mL of water to be drawn into the instrument
 - iv. Return the SIT arm to the close position.
 - d. **Go to the refrigerator next to the Fortessa**
 - i. Retrieve the blue cardboard CST box labeled “Fortessa”
 - e. If a prepped CST tube is already within the box and has at least 300uL continue, otherwise make a new CST sample.
 - i. New CST sample preparation
 1. Vortex the CST bead stock in all 4 orientations (bottom, sideways, upside down, sideways) to fully mix the stock solution.
 2. Place 500uL of DI water into a FACS tube (Just over the weld line at the bottom of the tube)
 3. Place one drop of CST bead stock into the tube
 4. Vortex the CST bead sample.
 - f. Load the CST sample onto the Fortessa SIP.
 - g. Obtain the current CST bead lot from the CST beads stock tube (do NOT use the lot# on the box)
 - i. Confirm/Select the proper bead lot from the dropdown – right side of CST window
 - h. **Go to the physical Fortessa control panel** (front of instrument)
 - i. Turn [Fine Adjustment] knob counter-clockwise until it stops (can take up to 10 full turns) and then turn it 5 full turns clockwise – to place potentiometer into the middle of its range.
 - ii. Press the instrument [Run] button.
 - i. **Return to Fortessa computer and the CST software window**
 - i. Press “Run” - in the CST software to start QC. *note: CST will take about 5mins
 - j. Once CST is completed
 - i. Remove the CST beads sample from the SIT

- ii. Put a tube of 2mL DI water on the SIT
- iii. Cap the CST sample tube and return to the CST box.
- iv. Return the CST box to the refrigerator
- v. Click “view report” in popup window to bring up CST report. (Review the report every run!!)
 - 1. Remember that issues with CST are important to look out for... but ultimately the most important factor is population identification. Errors on channels you are not using are much less of a concern than those you are using.
- vi. In the CST report window/table:
 - 1. Find bright bead robust CV column and check each primary channel (Blue B, Red C, Violet F, UV B, Yellow Green E)
 - a. If CV less than 4.5% - pass no attention needed
 - b. If CV is greater than 4.5% but less than 6.0% make a note and email JVD
 - c. If CV is 6.0% or greater try a new bead preparation (unless you made them new today), if this does not solve the issue conduct an abridged cleaning procedure – 10% bleach for 5mins, DI water for 5mins – rerun CST. If problem persists find JVD, do not be afraid to still try samples!
 - 2. Check Δ PMTV (Delta PMT Voltage) for ALL detectors
 - a. If Δ PMTV less than 30+/- pass no attention needed
 - b. If Δ PMTV is greater than 30+/- but less than 50+/- make a note and email JVD
 - c. If Δ PMTV is 50+/- or greater try a new bead preparation (unless you made them new today), if this does not solve the issue conduct an abridged cleaning procedure – 10% bleach for 5mins, DI water for 5mins – rerun CST. If problem persists find JVD, do not be afraid to still try samples!
 - 3. Check laser delay – this is critical on the Fortessa!
 - a. Laser delay on FSC for the yellow/Green laser should be +/- 5 the value reported on the laser delay card above Fortessa computer. (subject to change due to alignment changes)
 - b. This is critical for determining the stability of the fluidic subsystem.
- 8. Exit CST
- 9. Move to the **DIVA use SOP**.